

GradeSense — Graded Answer Paper

Question 1 — Electric Circuit Principles

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2.5 / 5

Evaluated

100% confidence

GradeSense - Answer Sheet

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Q1. Science (5 marks)

An electric circuit is a close path in which electric current can flow. The battery gives

the

potential difference which pushes the current around the circuit. Switch is used to open and close the circuit, if switch is closed then current will flow and bulb will glow, if switch is

open then no current flows through the circuit.

In the circuit the battery, switch, resistor, bulb and ammeter are connected in series because

current passes through all of them one by one. Ammeter is connected in series because it measures

the current flowing in the circuit. Voltmeter is also connected in the circuit to measure the potential difference, as shown in the diagram below -

When current passes through the circuit some of the current gets used up by the bulb and resistor,

that is why the bulb glows and becomes hot.

As per Ohm's law, $V = IR$. So if we increase the resistance then the current flowing in the circuit will also increase, because more resistance needs more current to push through it.

If

the resistance is less then the bulb will glow dim.

1

2

3

Rubric Breakdown

Score: 2.5 / 5

■ **Main circuit components (battery, switch, bulb, resistor) connected correctly in series** 1/1

Correctly identifies that the main components (battery, switch, resistor, bulb) are connected in series.

■ **[2] Ammeter connected in series and voltmeter connected in parallel across the bulb (or component being** 0.5/1

Correctly states the ammeter is connected in series. However, it does not specify that the voltmeter must be connected in parallel across the component whose voltage is being measured.

■ **Explains closed path of current flow and functional roles of components** 1/1

Clearly explains the concept of a closed path for current flow and the functional roles of the battery and switch.

■ **[3] Explains Ohm's law relationship (increased resistance decreases current flowing when voltage is constant)** 0/1

While Ohm's Law ($V=IR$) is stated, the explanation of its relationship between resistance and current is incorrect. Increasing resistance *decreases* current for a constant voltage, and vice-versa. The statement 'If the resistance is less then the bulb will glow dim' is also incorrect.

■ **[1] Clear, logically structured explanation with a properly labelled diagram and conventional current direction** 0/1

AI visual assessment (from the uploaded page image): While the diagram is properly labelled and indicates conventional current direction, the explanation contains significant conceptual errors, such as stating current is 'used up' and incorrectly describing the relationship between resistance and current (e.g., 'if we increase the resistance then the current flowing in the circuit will also increase'). Therefore, the explanation is not clear or logically structured from a physics standpoint.

Model Answer — Reference

Not graded — provided as the marking-scheme reference for this question.

A simple electric circuit provides a closed path through which electric current can flow. The battery provides the potential difference that drives current through the circuit. The switch is used to open or close the circuit. When the switch is closed, there is a complete conducting path and current can flow through the bulb and resistor. When the switch is open, the path is broken and current does not flow.??The bulb, resistor, battery, switch and ammeter should be connected in series in the main circuit. The ammeter must be connected in series because it measures the current flowing through the circuit. A voltmeter, however, should be connected in parallel across the bulb, because it measures the potential difference between the two ends of the bulb.??A properly labelled diagram would therefore show the battery connected to the switch, resistor, bulb and ammeter in a single closed loop, with the voltmeter connected across the bulb. The conventional current direction should be shown from the positive terminal of the battery through the external circuit towards the negative terminal.??The amount of current flowing through the circuit depends on both the potential difference and the resistance. According to Ohm's law, $V = IR$. If the voltage of the battery remains constant and the resistance is increased, the current flowing through the circuit decreases. Conversely, reducing the resistance allows more current to flow.